

# INTRODUCTION TO NETWORKS

## *Assignment Tasks & Solutions*

### Task 1: Network Path Diagram: Phone to Instagram

A conceptual breakdown of the physical and wireless connections established when a smartphone accesses Instagram:

- **1. Smartphone (Client Device):** Your mobile phone running the Instagram app initiates the request via built-in Wi-Fi or cellular radio.
- **2. Wireless Connection (Wi-Fi / Air):** Radio frequency signals (IEEE 802.11) carry the encrypted data packets from your phone through the air to your local wireless router.
- **3. Wi-Fi Router (Local Gateway):** The router receives the wireless frames, translates them, and forwards them across a wired Ethernet cable to the modem.
- **4. ISP (Internet Service Provider):** Your ISP takes the local traffic and routes it across their high-speed backbone fiber-optic network (WAN) toward the global internet infrastructure.
- **5. Instagram Server (Data Center):** The request traverses multiple intermediate routers and undersea/underground cables until it reaches Instagram's cloud servers, which process the request and send back the response.

### Task 2: Daily Network Devices & Network Types

Below are 5 common devices used daily and the specific type of network they primarily connect to:

| Device       | Network Type     | Description / Explanation                                                                                                         |
|--------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Smartphone   | PAN / WLAN / WAN | Connects via Bluetooth (PAN) to wireless earbuds, Wi-Fi (WLAN) at home/college, and cellular towers (WAN) on the move.            |
| Smart TV     | WLAN (LAN)       | Connects locally to the home router via Wi-Fi or Ethernet cable to stream media from servers over the internet.                   |
| Wi-Fi Router | LAN / WAN        | Acts as the central bridging hub for the Local Area Network (LAN) and interfaces directly with the ISP's Wide Area Network (WAN). |
| Laptop       | LAN / WLAN       | Connects to local office or college networks using Ethernet cables (LAN) or secure campus Wi-Fi (WLAN).                           |

|            |     |                                                                                                                                          |
|------------|-----|------------------------------------------------------------------------------------------------------------------------------------------|
| Smartwatch | PAN | Connects directly to your smartphone via Bluetooth Low Energy to sync health metrics and notifications within a 10-meter personal space. |
|------------|-----|------------------------------------------------------------------------------------------------------------------------------------------|

### Task 3: Difference Between the Internet and an Intranet

To understand the fundamental distinction between the public internet and a private intranet, we can examine a typical college campus environment:

- **The Internet:** The internet is a globally connected public web of computer networks accessible to anyone with an active connection. At college, when students browse external websites like YouTube, Wikipedia, or check personal email Gmail accounts, they are utilizing the public internet. It operates with open, global standards and has no single centralized owner.
- **An Intranet:** An intranet is a private, restricted network bounded securely within an organization. For example, a college student portal or internal LMS (Learning Management System) hosted on campus servers can only be accessed by enrolled students and faculty (often restricted behind campus Wi-Fi or VPN). External users on the public internet cannot access or view these internal academic records or administrative databases.

### Task 4: Data Journey During an App Search (WhatsApp Example)

Tracing the path of data packets when performing a search query inside a mobile application:

*When you type a contact name into the WhatsApp search bar and hit enter, your phone converts the keystrokes into digital data packets. These packets travel via your local Wi-Fi router through your ISP's broadband network across the public internet (WAN) into WhatsApp's secure cloud data center. WhatsApp's servers query their database to match the search query and immediately transmit the results back along the exact reverse path. Within milliseconds, your phone receives the incoming data packets and instantly renders the matching contact list on your screen.*